Matthias Heinz

Research Scientist

RIKEN Nishina Center
matthias.heinz@riken.jp
ORCID: 0000-0002-6363-0056

Education

Dr. rer. nat.	Technische Universität Darmstadt	2024
M.Sc.	Technische Universität Darmstadt	2020
B.Sc.	The Ohio State University	2018

Appointments

Research Scientist	RIKEN Nishina Center	2026–present
Eugene P. Wigner Fellow	Oak Ridge National Laboratory	2024–2026

Fellowships and Awards

FRIB Achievement Award for Early Career Researchers (Theory)	Facility for Rare Isotope Beams	2026
Distinguished Staff Fellowship	Oak Ridge National Laboratory	2024–2026
Giersch Award for an Outstanding Doctoral Thesis	Giersch Foundation	2024
Gerhard Herzberg Research Prize	Gerhard Herzberg Society	2024
Giersch Excellence Award	Giersch Foundation	2022
M.Sc. Research Prize	Gerhard Herzberg Society	2022
Doctoral Fellowship	German Academic Scholarship Foundation	2022–2024
Merit pregraduation scholarship for international students	Technische Universität Darmstadt	2020
Division of Nuclear Physics CEU Travel and Lodging Award	Division of Nuclear Physics	2015–2017
Physics Summer Research Scholarship	The Ohio State University	2015
Maximus Scholarship	The Ohio State University	2013–2017

Referee

Physical Review Letters, Physics Letters B, Physical Review C, Computer Physics Communications, The European Physical Journal A, Journal of Physics G, Physica Scripta

National Science Foundation Quantum Information Science program

ORNL Director's Research and Development program, ASCR Leadership Computing Challenge (ALCC), INCITE (Innovative and Novel Computational Impact on Theory and Experiment) program, Oak Ridge Leadership Computing Facility Director's Discretion program

Professional Societies

American Physical Society, Deutsche Physikalische Gesellschaft, Phi Beta Kappa

Funding

AI/ML-Ready Data Labeling for Low Energy Nuclear Physics (PI: Mario Cromaz)	$2,000,000	2025–2026
STREAMLINE 2: Machine Learning for Nuclear Many-Body Systems (PI: Dean Lee)	$1,370,000	2025–2027
ORNL LDRD Grant (single PI)	$603,800	2024–2027

Computing Grants

Jülich Supercomputing Centre (GCS project scalannm)	CPU: 4.3M core-h, GPU: 1.3M core-h (2026)	2024–present
Lichtenberg II Cluster (project 2455)	CPU: 0.3M core-h	2024
Jülich Supercomputing Centre (GCS project tensornuclei)	CPU: 7.5M core-h, GPU: 2.8M core-h	2021–2024
Lichtenberg II Cluster (project 1949)	CPU: 1.0M core-h	2022

Students Advised

Vivek Booshan (ongoing), Ben Johnson-Toth (summer 2025), Aidan O'Bryan (summer 2026), Tom Plies (B.Sc. 2022, M.Sc. 2024), Maxwell Rothman (ongoing)

Lectures at Physics Schools

Nuclear Physics Tortoise Lecture Series (Seoul, South Korea): Ten 75-minute lectures and five 90-minute exercises on "Quantum computing in nuclear physics"	2027
Exotic Beam Summer School (Oak Ridge, TN, USA): Two 60-minute lectures on "Modern nuclear structure theory"	2026
Multi-Particle Summer School Berkeley (Berkeley, CA, USA): Five 40-minute lectures on "Ab initio computations of nuclei"	2025
Nuclear Physics Turtle Lecture Series (Busan, South Korea): Six 75-minute lectures and three 90-minute exercises on "Ab initio Hartree-Fock calculations of nuclei"	2025
Arbeitstreffen Kernphysik (Schleching, Germany): Three 75-minute lectures on "Modern Nuclear Structure"	2023

Workshops Co-organized

Oak Ridge Annual Nuclear Gathering Event 2026	2026
RadIs Workshop: "Radioactive Isotopes for Fundamental Physics"	2026
ECT* Workshop: "Nuclear theory for astrophysics: Bridging bound-state and continuum physics"	2026
Oak Ridge Annual Nuclear Gathering Event 2025	2025
CRC 1245 Mini-Workshop: "Neutrino-nucleus interactions in the standard model and beyond"	2024

Research

Matthias Heinz researches the theoretical description of atomic nuclei. His specific interests are in understanding nuclear forces, simulating how the structure and dynamics of nuclei emerge from nuclear forces, and

predicting how the structure of nuclei impacts fundamental processes in nature, including the synthesis of heavy elements and possible new physics processes beyond the standard model. Matthias employs effective field theories, ab initio quantum many-body methods, Bayesian statistics, and emulators based on machine learning techniques in his research. Matthias is also exploring how quantum computing can accelerate quantum many-body simulations.

Author of 31 papers (21 published) with 579 citations on Google Scholar (October 31, 2026). Average citations per paper: 18.7. h-index: 11 on Google Scholar, 11 on INSPIRE. 47 invited talks, seminars, and colloquia.

Selected Publications

- *Toward scalable quantum computations of atomic nuclei*
  C. Gu, **M. Heinz**, O. Kiss, T. Papenbrock, Physical Review C **113**, 034321 (2026), arXiv:2507.14690
- *NuLattice: Ab initio computations of atomic nuclei on lattices*
  M. Rothman, B. Johnson-Toth, G. Hagen, **M. Heinz**, T. Papenbrock, The European Physical Journal A **62**, 28 (2026), arXiv:2509.08771
- *Ab initio calculations of overlap integrals for $\mu \to e$ conversion in nuclei*
  M. Heinz, M. Hoferichter, T. Miyagi, F. Noël, A. Schwenk, Physics Letters B **871**, 139975 (2025), arXiv:2412.04545
- *Improved structure of calcium isotopes from ab initio calculations*
  M. Heinz, T. Miyagi, S. R. Stroberg, A. Tichai, K. Hebeler, A. Schwenk, Physical Review C **111**, 034311 (2025), arXiv:2411.16014
- *Probing new bosons and nuclear structure with ytterbium isotope shifts*
  M. Door, C.-H. Yeh, **M. Heinz**, F. Kirk, C. Lyu, *et al.*, Physical Review Letters **134**, 063002 (2025), arXiv:2403.07792
- *In-medium similarity renormalization group with three-body operators*
  M. Heinz, A. Tichai, J. Hoppe, K. Hebeler, A. Schwenk, Physical Review C **103**, 044318 (2021), arXiv:2102.11172

A complete list of publications can be found below, on INSPIRE, and on Google Scholar.

Publications

1. *Beta decay of potassium-54 reveals the role of two-body currents in neutron-rich nuclei*
   Z. Y. Xu, R. K. Grzywacz, **M. Heinz**, T. Papenbrock, W. Ong, J. Allmond, B. C. Rasco, K. P. Rykaczewski, H. Schatz, B. M. Sherrill, O. Tarasov, P. Gysbers, G. Hagen, P. Navrátil, S. Ajayi, H. Arora, A. D. Ayangeakaa, H. Berg, J. Berkman, D. Bleuel, K. Bosmpotinis, N. Braukman, M. P. Carpenter, G. Cerizza, A. Chester, J. Christie, I. Cox, H. L. Crawford, B. Crider, J. Davis, A. Doetsch, J. Garcia-Duarte, A. Estrade, A. Fijalkowska, C. Frantzis, K. Fukushima, T. Gaballah, T. Gray, E. Good, K. Haak, S. Hanai, J. Harke, C. Harris, A. Hartley, M. Hausmann, K. Hermansen, D. Hoff, D. Hoskins, J. Huffman, R. Jain, M. Karny, T. King, N. Kitamura, K. Kolos, E. Kwan, S. N. Liddick, B. Longfellow, R. S. Lubna, S. Lyons, M. Madurga, M. Mogannam, S. Neupane, T. H. Ogunbeku, G. Owens-Fryar, J. Palomino, M. Portillo, M. M. Rajabali, A. Richard, I. Richardson, E. Ronning, G. Rose, T. Ruland, N. D. Scielzo, D. P. Scriven, D. Seweryniak, K. Siegl, M. Smith, A. Spyrou, M. Stepaniuk, A. Sweet, V. Tripathi, A. Tsantiri, S. Uthayakumaar, W. Walters, S. Watters, M. Wolinska-Cichocka, R. Yokoyama, Nature Communications (in press), arXiv:2610.36218

2. *Laser spectroscopy illuminates the $N = 32$ shell closure*
   T. E. Lellinger, L. V. Rodríguez, P. Müller, O. Ahmad, M. L. Bissell, K. Blaum, E. Burbach, B. Cheal, T. Fabritz, R. F. Garcia Ruiz, **M. Heinz**, J. Hughes, P. Imgram, K. König, Y. Liu, B. Maaß, E. N. Matthews, T. Miyagi, W. Nazarewicz, R. Neugart, G. Neyens, L. Nies, W. Nörtershäuser, J. Palmes, P. Plattner, P.-G. Reinhard, L. Renth, R. Sanchez, A. Schwenk, J. Spahn, X. F. Yang, D. T. Yordanov, Nature Communications (in press), arXiv:2608.10943

3. *Similarity renormalization group for nuclear forces*
   M. Heinz, Encyclopedia of Nuclear Physics (in press), arXiv:2603.04311

4. *Accurate charge radius measurement of ^{14}C confronts ab initio theory*
   K. König, P. Müller, T. Gesser, E. Burbach, S. Gandolfi, **M. Heinz**, P. Imgram, A. Lovato, P. Maris, T. Miyagi, W. Nörtershäuser, R. Roth, A. Schwenk, J. Spahn, Physical Review Letters **137**, 182501 (2026), arXiv:2608.17052

5. *Uncertainties with low-resolution nuclear forces*
   T. Plies, **M. Heinz**, A. Schwenk, Physical Review C **114**, 024327 (2026), arXiv:2509.24671

6. *Computational schemes for the Magnus expansion of the in-medium similarity renormalization group*
   M. Heinz, Physical Review C **113**, L041301 (2026), arXiv:2601.16133

7. *Toward scalable quantum computations of atomic nuclei*
   C. Gu, **M. Heinz**, O. Kiss, T. Papenbrock, Physical Review C **113**, 034321 (2026), arXiv:2507.14690
   Editors' Suggestion

8. *NuLattice: Ab initio computations of atomic nuclei on lattices*
   M. Rothman, B. Johnson-Toth, G. Hagen, **M. Heinz**, T. Papenbrock, The European Physical Journal A **62**, 28 (2026), arXiv:2509.08771

9. *Ab initio computations of the fourth-order charge density moments of ^{48}Ca and ^{208}Pb*
   T. Miyagi, **M. Heinz**, A. Schwenk, Physics Letters B **872**, 140032 (2026), arXiv:2508.10767

10. *Exactness of the normal-ordered two-body truncation of three-nucleon forces*
    M. Rothman, B. Johnson-Toth, F. Bonaiti, G. Hagen, **M. Heinz**, T. Papenbrock, Physical Review C **112**, L051301 (2025), arXiv:2508.01507

11. *Ab initio calculations of overlap integrals for $\mu \to e$ conversion in nuclei*
    M. Heinz, M. Hoferichter, T. Miyagi, F. Noël, A. Schwenk, Physics Letters B **871**, 139975 (2025), arXiv:2412.04545

12. *Connecting relativistic density functional theory to microscopic calculations*
B. T. Reed, **M. Heinz**, P. Arthuis, A. Schwenk, I. Tews, Physical Review C **112**, 034331 (2025), arXiv:2505.00828

13. *The nuclear charge radius of* ^{13}C
P. Müller, **M. Heinz**, P. Imgram, K. König, B. Maaß, T. Miyagi, W. Nörtershäuser, R. Roth, A. Schwenk, Nature Communications **16**, 6234 (2025), arXiv:2507.05680

Press Release: "How large is a carbon nucleus?"

14. *Improved structure of calcium isotopes from ab initio calculations*
M. Heinz, T. Miyagi, S. R. Stroberg, A. Tichai, K. Hebeler, A. Schwenk, Physical Review C **111**, 034311 (2025), arXiv:2411.16014

15. *Probing new bosons and nuclear structure with ytterbium isotope shifts*
M. Door, C.-H. Yeh, **M. Heinz**, F. Kirk, C. Lyu, T. Miyagi, J. C. Berengut, J. Bieroń, K. Blaum, L. S. Dreissen, S. Eliseev, P. Filianin, M. Filzinger, E. Fuchs, H. A. Fürst, G. Gaigalas, Z. Harman, J. Herkenhoff, N. Huntemann, C. H. Keitel, K. Kromer, D. Lange, A. Rischka, Ch. Schweiger, A. Schwenk, N. Shimizu, T. E. Mehlstäubler, Physical Review Letters **134**, 063002 (2025), arXiv:2403.07792

Press Release: "New insights into the structure of atomic nuclei"
Highlighted in Physics World: "Atomic anomaly explained without recourse to hypothetical 'dark force'"

16. *Randomized low-rank decompositions of nuclear three-body interactions*
A. Tichai, P. Arthuis, K. Hebeler, **M. Heinz**, J. Hoppe, T. Miyagi, A. Schwenk, L. Zurek, Physical Review Research **6**, 043331 (2024), arXiv:2307.15572

17. *Spectroscopy of* ^{52}K
M. Enciu, A. Obertelli, P. Doornenbal, **M. Heinz**, T. Miyagi, F. Nowacki, K. Ogata, A. Poves, A. Schwenk, K. Yoshida, N. L. Achouri, H. Baba, F. Browne, D. Calvet, F. Château, S. Chen, N. Chiga, A. Corsi, M. L. Cortés, A. Delbart, J.-M. Gheller, A. Giganon, A. Gillibert, C. Hilaire, T. Isobe, T. Kobayashi, Y. Kubota, V. Lapoux, H. N. Liu, T. Motobayashi, I. Murray, H. Otsu, V. Panin, N. Paul, W. Rodríguez, H. Sakurai, M. Sasano, D. Steppenbeck, L. Stuhl, Y. L. Sun, Y. Togano, T. Uesaka, K. Wimmer, K. Yoneda, O. Aktas, T. Aumann, L. X. Chung, F. Flavigny, S. Franchoo, I. Gašparić, R.-B. Gerst, J. Gibelin, K. I. Hahn, D. Kim, Y. Kondo, P. Koseoglou, J. Lee, C. Lehr, P. J. Li, B. D. Linh, T. Lokotko, M. MacCormick, K. Moschner, T. Nakamura, S. Y. Park, D. M. Rossi, E. Sahin, P.-A. Söderström, D. Sohler, S. Takeuchi, H. Toernqvist, V. Vaquero, V. Wagner, S. Wang, V. Werner, X. Xu, H. Yamada, D. Yan, Z. H. Yang, M. Yasuda, L. Zanetti, Physical Review C **110**, 064301 (2024), arXiv:2412.03602

18. *Normal ordering of three-nucleon interactions for ab initio calculations of heavy nuclei*
K. Hebeler, V. Durant, J. Hoppe, **M. Heinz**, A. Schwenk, J. Simonis, A. Tichai, Physical Review C **107**, 024310 (2023), arXiv:2211.16262

19. *Perspectives on an open-source toolchain for ab initio nuclear physics*
M. Heinz, T. Duguet, H. W. Grießhammer, H. Hergert, A. Tichai, Contribution in "Nuclear forces for precision nuclear physics: A collection of perspectives" Few-Body Systems **63**, 67 (2022), arXiv:2202.01105

20. *Least-square approach for singular value decompositions of scattering problems*
A. Tichai, P. Arthuis, K. Hebeler, **M. Heinz**, J. Hoppe, A. Schwenk, L. Zurek, Physical Review C **106**, 024320 (2022), arXiv:2205.10087

21. *Importance truncation for the in-medium similarity renormalization group*
J. Hoppe, A. Tichai, **M. Heinz**, K. Hebeler, A. Schwenk, Physical Review C **105**, 034324 (2022), arXiv:2110.09390

22. *Low-rank matrix decompositions for ab initio nuclear structure*
A. Tichai, P. Arthuis, K. Hebeler, **M. Heinz**, J. Hoppe, A. Schwenk, Physics Letters B **821**, 136623 (2021), arXiv:2105.03935
23. *In-medium similarity renormalization group with three-body operators*
M. Heinz, A. Tichai, J. Hoppe, K. Hebeler, A. Schwenk, Physical Review C **103**, 044318 (2021), arXiv:2102.11172
24. *Natural orbitals for many-body expansion methods*
J. Hoppe, A. Tichai, **M. Heinz**, K. Hebeler, A. Schwenk, Physical Review C **103**, 014321 (2021), arXiv:2009.04701

Manuscripts in Review

1. *Searching for charged lepton flavor violation*
L. Calibbi, K. Fuyuto, **M. Heinz**, J. Zupan, arXiv:2610.11839
2. *Probing the size of neutron and proton single-particle orbitals from nucleon knockout reactions*
M. Enciu, A. Obertelli, P. Doornenbal, C. Barbieri, S. Brolli, **M. Heinz**, W. Horiuchi, T. Inakura, W. H. Long, T. Miyagi, F. Nowacki, K. Ogata, A. Poves, A. Schwenk, K. Yoshida, N. L. Achouri, H. Baba, F. Browne, D. Calvet, F. Château, S. Chen, N. Chiga, A. Corsi, M. L. Cortés, A. Delbart, J.-M. Gheller, A. Giganon, A. Gillibert, C. Hilaire, T. Isobe, T. Kobayashi, Y. Kubota, V. Lapoux, H. N. Liu, T. Motobayashi, I. Murray, H. Otsu, V. Panin, N. Paul, W. Rodríguez, H. Sakurai, M. Sasano, D. Steppenbeck, L. Stuhl, Y. L. Sun, Y. Togano, T. Uesaka, K. Wimmer, K. Yoneda, O. Aktas, T. Aumann, L. X. Chung, F. Flavigny, S. Franchoo, I. Gašparić, R.-B. Gerst, J. Gibelin, K. I. Hahn, D. Kim, Y. Kondo, P. Koseoglou, J. Lee, C. Lehr, P. J. Li, B. D. Linh, T. Lokotko, M. MacCormick, K. Moschner, T. Nakamura, S. Y. Park, D. M. Rossi, E. Sahin, P.-A. Söderström, D. Sohler, S. Takeuchi, H. Toernqvist, V. Vaquero, V. Wagner, S. Wang, V. Werner, X. Xu, H. Yamada, D. Yan, Z. H. Yang, M. Yasuda, L. Zanetti, arXiv:2608.16549
3. *Ab initio calculations of parity-violating electron scattering off* ^{48}Ca *and* ^{208}Pb
F. Noël, **M. Heinz**, M. Hoferichter, T. Miyagi, A. Schwenk, arXiv:2606.25019
4. *Saturation of nuclear binding from lattice Hamiltonians*
M. Rothman, G. Hagen, **M. Heinz**, T. Papenbrock, arXiv:2606.12166
5. *Taming nuclear size and shape effects in superallowed beta-decay*
B. He, M. Gorchtein, **M. Heinz**, B. Ohayon, L. Platter, C.-Y. Seng, arXiv:2605.13985
6. *Nuclear charge radii of aluminium isotopes at the proton drip line*
A. Brinson, B. Rickey, P. Arthuis, A. Belley, S. Campbell, X. Chen, A. Dockery, S. Elhatisari, H. Erington, N. Gamage, R. F. Garcia Ruiz, **M. Heinz**, J. Holt, C. Ireland, C. Izzo, C. Jones, J. Karthein, K. König, D. Lee, Y.-Z. Ma, F. Maier, U.-G. Meißner, K. Minamisono, M. Moenter, J. M. Munoz, W. Nörtershäuser, A. Ortiz-Cortes, J. Palmes, S. Papa, F. P. Cruz, R. Ringle, H. Sims, C. Sumithrarachchi, A. R. Vernon, T. Wang, S. G. Wilkins, R. Yadav, S. Zhang, arXiv:2605.09139
7. *First spectroscopy of* ^{100}*Sn establishes a robust* $Z = N = 50$ *shell closure*
X. Liu, M. L. Cortés, T. Gao, J. Lee, P. Doornenbal, F. Browne, S. Chen, D. D. Dao, Z. Elekes, G. Hagen, **M. Heinz**, R. Hu, S. Koyama, M. Kurata-Nishimura, Z. Li, W. Marshall, T. Miyagi, F. Nowacki, A. Obertelli, K. Ogata, H. Otsu, T. Otsuka, T. Papenbrock, S. Péru, T. Plies, A. Schwenk, N. Shimizu, D. Sohler, R. Taniuchi, Y. Tsunoda, K. Yoshida, C. Yuan, J. Acosta, A. Algora, M. Begala, C. Fu, Z. Halasz, Y. Hirayama, D. Hou, S. Iimura, H. Ishiyama, Y. Jang, M. Juhász, A. Jungclaus, S. Kimura, G. G. Kiss, I. Kuti, J. Lee, M. Li, Z. Li, H. Liu, B. Moon, S. Nishimura, T. Niwase, J. Park, S. Paschalis, M. Petri, V. Phong, M. Rosenbusch, H. Sakurai, P. Schury, T. Sonoda, A. Takamine, L. Tetley, M. Wada, H. Wang, S. Wang, S. Wang, Y. Watanabe, W. Xian, W. Xu, Y. Xu, Y. Yang, J. M. Yap, Y. Yu, Q. Zeng, S. Zha, K. Zheng

Presentations at Conferences and Workshops

1. "NuLattice: Computations of nuclei on lattices"
   CRC 1245 Workshop
   Mainz, Germany — September 2026
   Invited Talk
2. "Improving precision and reach of ab initio nuclear theory"
   Low Energy Community Meeting 2026
   Argonne National Laboratory, Lemont, IL, USA — August 2026
   Plenary Talk: *FRIB Achievement Award for Early Career Researchers*
3. "Ab initio theory for nuclear densities and parity-violating electron scattering"
   Nuclear, Atomic, Molecular, and Optical Physics Workshop
   Argonne National Laboratory, Lemont, IL, USA — August 2026
   Invited Talk
4. "NuLattice: Computations of nuclei on lattices"
   SciDAC PI Meeting 2026
   Rockville, MD, USA — August 2026
   Invited Talk
5. "Computations of parity-violating electron scattering off ^{48}Ca and ^{208}Pb"
   SciDAC PI Meeting 2026
   Rockville, MD, USA — August 2026
   Poster
6. "NuLattice: Computations of nuclei on lattices"
   International Symposium on Lattice Field Theory (Lattice 2026)
   University of Maryland, College Park, MD, USA — July 2026
   Contributed Talk
7. "Computing Nuclei from First Principles: Quantum Computations, AI Emulation, Deformation, and Fundamental Symmetries"
   Atoms to Applications Symposium
   Oak Ridge National Laboratory, Oak Ridge, TN, USA — June 2026
   Poster
8. "Hamiltonians for neutron densities and lattice calculations"
   INT Program: Nuclear Hamiltonians for Advancing Nuclear Physics and Beyond
   Institute for Nuclear Theory, Seattle, WA, USA — April 2026
   Invited Talk
9. "Nuclear radii from ab initio calculations: Successes and challenges"
   Workshop: Precise nuclear charge radii and beyond
   Max Planck Institute for Nuclear Physics, Heidelberg, Germany — January 2026
   Invited Talk
10. "Precise neutron densities of nuclei and what this means for neutron skins"
    Hirschegg 2026 – Challenges in effective field theory descriptions of nuclei
    Hirschegg, Austria — January 2026
    Invited Talk
11. "Nuclear densities, neutron skins, and IMSRG on the lattice"
    IMSRG Workshop: Flow and Tell
    University of Notre Dame, Notre Dame, IN, USA — October 2025
    Invited Talk
12. "Leadership computing for atomic nuclei and fundamental physics"
    Smoky Mountains Computational Sciences and Engineering Conference
    Chattanooga, TN, USA — August 2025
    Poster

13. "Developing precise predictions of electroweak structure of nuclei"
Workshop on Precision Measurements with Radioisotopes
Texas A&M University, College Station, TX, USA August 2025
Invited Talk
14. "First-principles nuclear structure theory and implications for fundamental physics"
ESNT Workshop: "Theoretical and experimental developments for symmetry-violating nuclear properties"
CEA Paris-Saclay, Saclay, France June 2025
Invited Talk
15. "Improving reach and precision of ab initio nuclear structure predictions"
Gordon Research Conference on Nuclear Chemistry
Colby-Sawyer College, New London, NH, USA June 2025
Invited Talk
16. "Exploring new physics and nuclear structure with ytterbium isotope shifts"
Conference on the Intersection of Nuclear and Particle Physics 2025
University of Wisconsin, Madison, WI, USA June 2025
Invited Parallel Talk
17. "Ab initio overlap integrals for muon to electron conversion in nuclei"
Conference on the Intersection of Nuclear and Particle Physics 2025
University of Wisconsin, Madison, WI, USA June 2025
Invited Parallel Talk
18. "Ab initio overlap integrals for mu to e conversion"
NUCLEI Collaboration Meeting 2025
University of North Carolina, Chapel Hill, NC, USA June 2025
Contributed Talk
19. "Ab initio nuclear structure for searches for $\mu \to e$ conversion"
ECT Workshop on "Lepton flavor change in nuclei"*
ECT*, Trento, Italy April 2025
Invited Talk
20. "Perturbative treatment of subleading interactions in ab initio calculations"
INT Workshop on "Chiral EFT: New Perspectives"
Institute for Nuclear Theory, Seattle, WA, USA March 2025
Invited Talk
21. "Ab initio nuclear structure for searches for $\mu \to e$ conversion"
Progress in Ab Initio Nuclear Theory 2025
TRIUMF, Vancouver, Canada February 2025
Invited Talk
22. "Probing new physics with ytterbium isotope shifts"
CERN Workshop on "Nuclear Shapes and BSM Searches at Colliders"
CERN, Geneva, Switzerland (virtual) January 2025
Invited Talk
23. "Ab initio nuclear theory for new physics searches in ytterbium and beyond"
ORNL Radioactive Molecules Workshop
Oak Ridge National Laboratory, Oak Ridge, TN, USA September 2024
Invited Talk
24. "Ab initio nuclear structure calculations for new physics searches in ytterbium isotope shifts"
Quark Confinement and the Hadron Spectrum Conference 2024
Cairns, Australia August 2024
Invited Parallel Talk

25. "Improved ab initio description of neutron-rich calcium isotopes"
Direct Reactions in Exotic Beams 2024
Wiesbaden, Germany June 2024
Poster: *1st Place Theory Poster Award*

26. "Precision ab initio structure and implications for fundamental physics"
Spring Meeting of the Deutsche Physikalische Gesellschaft
Gießen, Germany March 2024
Contributed Talk

27. "Improved medium-mass nuclear structure and responses with the IMSRG"
Progress in Ab Initio Nuclear Theory 2024
TRIUMF, Vancouver, Canada February 2024
Invited Talk

28. "IMSRG developments and medium-mass and heavy nuclei"
CRC 1245 Workshop
Technische Universität Darmstadt, Darmstadt, Germany September 2023
Invited Talk

29. "Diagrammatic resummations for the in-medium similarity renormalization group"
ESNT Workshop "Automated tools for many-body theory"
CEA Paris-Saclay, Saclay, France June 2023
Invited Talk

30. "Ab initio nuclear structure with the in-medium similarity renormalization group"
Spring Doctoral Forum of the Studienstiftung
Berlin, Germany April 2023
Contributed Talk

31. "Ab initio structure of neutron-rich calcium isotopes"
Spring Meeting of the Deutsche Physikalische Gesellschaft
Dresden, Germany March 2023
Contributed Talk

32. "IMSRG with three-body operators"
Progress in Ab Initio Nuclear Theory 2023
TRIUMF, Vancouver, Canada March 2023
Contributed Talk

33. "Improved ab initio nuclear structure for medium-mass and heavy nuclei"
Arbeitstreffen Kernphysik 2023
Schleching, Germany February 2023
Invited Talk

34. "Recent progress on in-medium similarity renormalization group calculations with three-body operators"
Hirschegg 2023 – Effective field theories for nuclei and nuclear matter
Hirschegg, Austria January 2023
Contributed Talk

35. "Extension and acceleration of the in-medium similarity renormalization group"
Spring Meeting of the Deutsche Physikalische Gesellschaft
Mainz, Germany (virtual) March 2022
Contributed Talk

36. "Approximations and uncertainties in the in-medium similarity renormalization group"
ISNET 8
Facility for Rare Isotope Beams, East Lansing, MI, USA December 2021
Contributed Talk

37. "Recent developments in the in-medium similarity renormalization group"
    CRC 1245 Workshop
    Technische Universität Darmstadt, Darmstadt, Germany — March 2021
    Invited Talk
38. "Exploring alternative SRG generators in one dimension"
    Spring Meeting of the Deutsche Physikalische Gesellschaft
    Munich, Germany — March 2019
    Contributed Talk
39. "Understanding uncertainties in jet quenching in high-energy nucleus-nucleus collisions"
    Fall Meeting of the Division of Nuclear Physics
    Pittsburgh, PA, USA — October 2017
    Poster
40. "Exploration of similarity renormalization group generators in 1-dimensional potentials"
    Fall Meeting of the Division of Nuclear Physics
    Vancouver, Canada — October 2016
    Poster
41. "Exploration of similarity renormalization group generators in 1-dimensional potentials"
    Fall Meeting of the Division of Nuclear Physics
    Santa Fe, NM, USA — October 2015
    Poster

Invited Lectures at Summer/Winter Schools

1. Ten 75-minute lectures and five 90-minute programming exercises on "Quantum computing in nuclear physics"
   Nuclear Physics Tortoise Lecture Series 2027
   Seoul, South Korea — January 2027
2. Two 60-minute lectures on "Modern nuclear structure theory"
   Exotic Beam Summer School 2026
   Oak Ridge, TN, USA — June 2026
3. Five 40-minute lectures on "Ab initio computations of nuclei"
   Multi-Particle Summer School Berkeley 2025
   Berkeley, CA, USA — July 2025
4. Six 75-minute lectures and three 90-minute programming exercises on "Ab initio Hartree-Fock calculations of nuclei"
   Nuclear Physics Turtle Lecture Series 2025
   Busan, South Korea — January 2025
5. Three 75-minute main speaker lectures on "Modern Nuclear Structure"
   Arbeitstreffen Kernphysik 2023
   Schleching, Germany — February 2023

Invited Colloquia and Seminars

1. "Ab initio analysis of parity-violating electron scattering"
   University of Tennessee Nuclear Physics Seminar
   University of Tennessee, Knoxville, TN, USA — September 2026
2. "Towards precise ab initio predictions of neutron densities and skins"
   University of Tsukuba, Nuclear Physics Seminar
   University of Tsukuba, Tsukuba, Japan — March 2026

3. "Towards precise ab initio predictions of neutron densities and skins"
   Notre Dame Nuclear Physics Seminar
   University of Notre Dame, Notre Dame, IN, USA — March 2026
4. "Neutron radii and densities and skins, oh my!"
   University of Tennessee Nuclear Physics Seminar
   University of Tennessee, Knoxville, TN, USA — March 2026
5. "Improving reach and precision of ab initio theory for new physics studies"
   Nuclear and Chemical Sciences Division Seminar
   LLNL, Livermore, CA, USA — July 2025
6. "Nuclei on quantum computers"
   ORNL Physics Division Summer Lecture Series
   ORNL, Oak Ridge, TN, USA — July 2025
7. "Improving reach and precision of ab initio theory for new physics studies"
   ANL PHY Division Theory Seminar
   Argonne National Laboratory, Lemont, IL, USA — June 2025
8. "Improving reach and precision of ab initio theory for new physics studies"
   Ohio University, INPP Seminar
   Ohio University, Athens, OH, USA — April 2025
9. "New bosons in ytterbium isotope shifts? An ab initio nuclear theory perspective"
   University of Tennessee Nuclear Physics Seminar
   University of Tennessee, Knoxville, TN, USA — November 2024
10. "Improved ab initio nuclear structure of neutron-rich calcium isotopes"
    R2S2 Seminar
    Rising Researchers Seminar Series (virtual) — June 2024
11. "First-principles nuclear structure theory and implications for fundamental physics"
    TRIUMF NP Seminar
    TRIUMF, Vancouver, Canada — February 2024
12. "First-principles nuclear structure theory and implications for fundamental physics"
    S@INT Seminar
    Institute for Nuclear Theory, Seattle, WA, USA — February 2024
13. "First-principles nuclear structure theory and implications for fundamental physics"
    TRIUMF Interview Seminar
    TRIUMF, Vancouver, Canada — January 2024
14. "First-principles nuclear structure calculations and implications for fundamental physics"
    ORNL – Distinguished Staff Fellowship Program – NCCS Seminar
    Oak Ridge National Laboratory, Oak Ridge, TN, USA — December 2023
15. "First-principles nuclear structure theory and implications for fundamental physics"
    FRIB Theory Seminar
    FRIB, East Lansing, MI, USA — December 2023
16. "First-principles nuclear structure theory and implications for fundamental physics"
    Subatomic, High Energy and Plasma Physics Seminar
    Chalmers University of Technology, Göteborg, Sweden — November 2023
17. "First-principles nuclear structure theory and implications for fundamental physics"
    ITP Seminar
    Universität Bern, Bern, Switzerland — November 2023
18. "Precision nuclear structure with the in-medium similarity renormalization group"
    Physics Seminar
    Yonsei University, Seoul, South Korea — May 2023

19. "Precision nuclear structure with the in-medium similarity renormalization group"
    Triangle Nuclear Theory Colloquium
    North Carolina State University, Raleigh, NC, USA — March 2023
20. "Precision nuclear structure with the in-medium similarity renormalization group"
    LANL T-2 Nuclear Theory Seminar
    Los Alamos National Laboratory, Los Alamos, NM, USA — March 2023
21. "Recent progress in ab initio nuclear structure calculations with the in-medium similarity renormalization group"
    MPIK Seminar – Stored and Cooled Ions
    Max-Planck-Institut für Kernphysik, Heidelberg, Germany — November 2022
22. "Progress on IMSRG with 3-body operators"
    ORNL Nuclear Structure Theory Group Seminar
    Oak Ridge National Laboratory, Oak Ridge, TN, USA — August 2022
23. "Analysis of three-body effects in the IMSRG"
    ANL Physics Theory Group Seminar
    Argonne National Laboratory, Lemont, IL, USA — July 2022